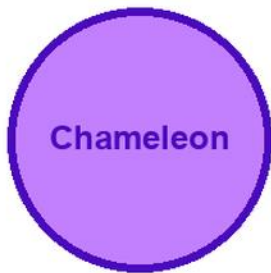


KITABU QUEST

Grade 9

SCIENCE AND TECHNOLOGY



Cover scene

learners observing a chameleon and plants during a safe outdoor investigation

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

9

KEN CBC aligned draft

Think - Explore - Achieve

Title Page

KITABU QUEST Grade 9 Science and Technology

Kenya CBC content draft for review

Mascot: chameleon

This content-first draft was regenerated from Kitabu curriculum data using a topic-by-topic book plan. It is designed to help learners read, practise, discuss, create, and revise with confidence.

Cover artwork is intentionally deferred until all content has passed review and is marked ready-for-cover.

How To Use This Book

This book helps Grade 9 learners study Science and Technology through clear lessons, worked examples, activities, practice, and revision.

Use the inquiry questions to think first. Read the explanation. Try the guided example. Complete the activity. Answer the practice questions. Correct your work and ask for support where needed.

Observe, ask questions, test safely, record evidence, and explain findings.

Every unit has a local example, a misconception to avoid, success criteria, and a check for understanding.

Learning Skills In This Book

Each chapter builds knowledge, skills, values, creativity, communication, collaboration, critical thinking, digital literacy, and self-confidence.

Look for these page types: Learn, Example, Activity, Practice, Reflection, Home Link, and Review.

How to study well: read the goal first, underline new words, try the example before reading the answer, discuss with a partner, and correct your work after feedback. Do not skip the reflection questions. They help you notice what you understand and what needs more practice.

Table Of Contents

Use this table of contents to plan your study. Start with the first chapter, but return to earlier pages whenever you need revision.

1. 1.0 Mixtures, Elements, and Compounds
2. 2.0 Living things and their Environment
3. 3.0 Force and Energy

After each chapter, complete the chapter review before moving forward. At the end of the book, use the glossary, answer notes, and final project to revise the whole subject.

Chapter: Mixtures, Elements, and Compounds

In this chapter you will study Mixtures, Elements, and Compounds.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 1.1 Structure of the atom
- 1.2 Metals and Alloys
- 1.3 Water hardness

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Structure of the atom: structure of the atom: Lesson Opener

Learning area: Structure of the atom: structure of the atom

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Structure of the atom: structure of the atom.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Structure of the atom: structure of the atom.

By the end of this lesson sequence, you should be able to:

- describe the structure of the atom
- determine the mass number of elements
- draw the electron arrangement in atoms using dot or cross diagrams

Inquiry questions:

- How is the structure of the atom important?

Before reading, write one thing you already know and one question you want this lesson to answer.

Structure of the atom: structure of the atom: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Structure of the atom: structure of the atom.

Important idea: Structure of the atom: structure of the atom.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Structure of the atom: structure of the atom
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Structure of the atom: structure of the atom
- Structure of the atom
- Structure
- atom

Explain the idea in your own words before attempting the activities.

Structure of the atom: structure of the atom: Worked Example

Model for Structure of the atom: structure of the atom: Connect subatomic particles to atom structure.

Question: How do particles inside an atom explain atomic number and mass number?

Example table:

Particle	Charge	Position
Proton	positive	nucleus
Neutron	neutral	nucleus
Electron	negative	shells around nucleus

Procedure:

1. Draw a nucleus and label protons and neutrons.
2. Draw electrons on shells around the nucleus.
3. State that atomic number equals number of protons.
4. State that mass number equals protons plus neutrons.
5. Use one example atom to calculate the missing value.

Conclusion frame: Atomic structure is useful because _____.

Structure of the atom: structure of the atom: Activity

Activity for Structure of the atom: structure of the atom: Plan a safe observation or investigation.

Inquiry question: How is the structure of the atom important?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Structure of the atom: structure of the atom: Activity And Practice

Practice for Structure of the atom: structure of the atom:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: describe the structure of the atom. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: determine the mass number of elements. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: draw the electron arrangement in atoms using dot or cross diagrams. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Structure of the atom: structure of the atom: Outcome Check

Outcome: describe the structure of the atom

Show mastery with topic evidence.

Task: Draw an atom model, label protons, neutrons, electrons, nucleus, and shells, then complete a small table for atomic number and mass number. Answer guide: protons are positive in the nucleus, neutrons are neutral in the nucleus, electrons are negative around the nucleus, atomic number equals protons.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Structure of the atom: structure of the atom, uses correct science vocabulary, and connects evidence to explanation.

Structure of the atom: structure of the atom: Outcome Check

Outcome: determine the mass number of elements

Show mastery with topic evidence.

Task: Draw an atom model, label protons, neutrons, electrons, nucleus, and shells, then complete a small table for atomic number and mass number. Answer guide: protons are positive in the nucleus, neutrons are neutral in the nucleus, electrons are negative around the nucleus, atomic number equals protons.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Structure of the atom: structure of the atom, uses correct science vocabulary, and connects evidence to explanation.

Structure of the atom: structure of the atom: Outcome Check

Outcome: draw the electron arrangement in atoms using dot or cross diagrams

Show mastery with topic evidence.

Task: Draw an atom model, label protons, neutrons, electrons, nucleus, and shells, then complete a small table for atomic number and mass number. Answer guide: protons are positive in the nucleus, neutrons are neutral in the nucleus, electrons are negative around the nucleus, atomic number equals protons.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Structure of the atom: structure of the atom, uses correct science vocabulary, and connects evidence to explanation.

Structure of the atom: classify elements into metals and non-metals: Lesson Opener

Learning area: Structure of the atom: classify elements into metals and non- metals

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Structure of the atom: classify elements into metals and non- metals.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Structure of the atom: classify elements into metals and non- metals.

By the end of this lesson sequence, you should be able to:

- classify elements into metals and non- metals
- show interest in classifying elements into metals and non- metals.

Inquiry questions:

- How is the structure of the atom important?

Before reading, write one thing you already know and one question you want this lesson to answer.

Structure of the atom: classify elements into metals and non-metals: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Structure of the atom: classify elements into metals and non- metals.

Important idea: Structure of the atom: classify elements into metals and non- metals.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Structure of the atom: classify elements into metals and non- metals
|
| --- | --- |
| Labelled sketch | Name visible parts and show direction or change with arrows. |
| Observation table | Record facts before explaining them. |
| Safety note | Name the hazard and the safe action. |
| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Structure of the atom: classify elements into metals and non- metals
- Structure of the atom
- Structure
- atom
- classify
- elements
- metals

Explain the idea in your own words before attempting the activities.

Structure of the atom: classify elements into metals and non-metals: Worked Example

Model for Structure of the atom: classify elements into metals and non- metals: Connect subatomic particles to atom structure.

Question: How do particles inside an atom explain atomic number and mass number?

Example table:

Particle	Charge	Position
Proton	positive	nucleus
Neutron	neutral	nucleus
Electron	negative	shells around nucleus

Procedure:

1. Draw a nucleus and label protons and neutrons.
2. Draw electrons on shells around the nucleus.
3. State that atomic number equals number of protons.
4. State that mass number equals protons plus neutrons.
5. Use one example atom to calculate the missing value.

Conclusion frame: Atomic structure is useful because _____.

Structure of the atom: classify elements into metals and non-metals: Activity

Activity for Structure of the atom: classify elements into metals and non- metals: Plan a safe observation or investigation.

Inquiry question: How is the structure of the atom important?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Structure of the atom: classify elements into metals and non-metals: Activity And Practice

Practice for Structure of the atom: classify elements into metals and non- metals:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: classify elements into metals and non- metals. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: show interest in classifying elements into metals and non- metals.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Structure of the atom: classify elements into metals and non-metals: Outcome Check

Outcome: classify elements into metals and non- metals

Show mastery with topic evidence.

Task: Draw an atom model, label protons, neutrons, electrons, nucleus, and shells, then complete a small table for atomic number and mass number. Answer guide: protons are positive in the nucleus, neutrons are neutral in the nucleus, electrons are negative around the nucleus, atomic number equals protons.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Structure of the atom: classify elements into metals and non- metals, uses correct science vocabulary, and connects evidence to explanation.

Structure of the atom: classify elements into metals and non-metals: Outcome Check

Outcome: show interest in classifying elements into metals and non- metals.

Show mastery with topic evidence.

Task: Draw an atom model, label protons, neutrons, electrons, nucleus, and shells, then complete a small table for atomic number and mass number. Answer guide: protons are positive in the nucleus, neutrons are neutral in the nucleus, electrons are negative around the nucleus, atomic number equals protons.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Structure of the atom: classify elements into metals and non- metals, uses correct science vocabulary, and connects evidence to explanation.

Metals and Alloys: physical properties of metals: Lesson Opener

Learning area: Metals and Alloys: physical properties of metals

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Metals and Alloys: physical properties of metals.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Metals and Alloys: physical properties of metals.

By the end of this lesson sequence, you should be able to:

- describe the physical properties of metals
- describe the composition of alloys
- identify the uses of metals and alloys in day-to-day life

Inquiry questions:

- How are alloys important in day-to-day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Metals and Alloys: physical properties of metals: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Metals and Alloys: physical properties of metals.

Important idea: Metals and Alloys: physical properties of metals.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Metals and Alloys: physical properties of metals |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Metals and Alloys: physical properties of metals
- Metals and Alloys
- Metals
- Alloys
- physical
- properties

Explain the idea in your own words before attempting the activities.

Metals and Alloys: physical properties of metals: Worked Example

Investigation model for Metals and Alloys: physical properties of metals: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Metals and Alloys: physical properties of metals: Activity

Activity for Metals and Alloys: physical properties of metals: Plan a safe observation or investigation.

Inquiry question: How are alloys important in day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Metals and Alloys: physical properties of metals: Activity And Practice

Practice for Metals and Alloys: physical properties of metals:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: describe the physical properties of metals. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: describe the composition of alloys. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: identify the uses of metals and alloys in day-to-day life. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Metals and Alloys: physical properties of metals: Outcome Check

Outcome: describe the physical properties of metals

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Metals and Alloys: physical properties of metals, uses correct science vocabulary, and connects evidence to explanation.

Metals and Alloys: physical properties of metals: Outcome Check

Outcome: describe the composition of alloys

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Metals and Alloys: physical properties of metals, uses correct science vocabulary, and connects evidence to explanation.

Metals and Alloys: physical properties of metals: Outcome Check

Outcome: identify the uses of metals and alloys in day-to-day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Metals and Alloys: physical properties of metals, uses correct science vocabulary, and connects evidence to explanation.

Metals and Alloys: explain the effects of rusting of metals: Lesson Opener

Learning area: Metals and Alloys: explain the effects of rusting of metals

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Metals and Alloys: explain the effects of rusting of metals.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Metals and Alloys: explain the effects of rusting of metals.

By the end of this lesson sequence, you should be able to:

- explain the effects of rusting of metals
- appreciate the importance of common alloys in day-to-day life.

Inquiry questions:

- How are alloys important in day-to-day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Metals and Alloys: explain the effects of rusting of metals: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Metals and Alloys: explain the effects of rusting of metals.

Important idea: Metals and Alloys: explain the effects of rusting of metals.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Metals and Alloys: explain the effects of rusting of metals
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Metals and Alloys: explain the effects of rusting of metals
- Metals and Alloys
- Metals
- Alloys
- explain
- effects

Explain the idea in your own words before attempting the activities.

Metals and Alloys: explain the effects of rusting of metals: Worked Example

Investigation model for Metals and Alloys: explain the effects of rusting of metals: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Metals and Alloys: explain the effects of rusting of metals: Activity

Activity for Metals and Alloys: explain the effects of rusting of metals: Plan a safe observation or investigation.

Inquiry question: How are alloys important in day-to- day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Metals and Alloys: explain the effects of rusting of metals: Activity And Practice

Practice for Metals and Alloys: explain the effects of rusting of metals:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: explain the effects of rusting of metals. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: appreciate the importance of common alloys in day-to-day life.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Metals and Alloys: explain the effects of rusting of metals: Outcome Check

Outcome: explain the effects of rusting of metals

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Metals and Alloys: explain the effects of rusting of metals, uses correct science vocabulary, and connects evidence to explanation.

Metals and Alloys: explain the effects of rusting of metals: Outcome Check

Outcome: appreciate the importance of common alloys in day-to-day life.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Metals and Alloys: explain the effects of rusting of metals, uses correct science vocabulary, and connects evidence to explanation.

Water hardness: physical properties of water: Lesson Opener

Learning area: Water hardness: physical properties of water

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Water hardness: physical properties of water.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Water hardness: physical properties of water.

By the end of this lesson sequence, you should be able to:

- describe the physical properties of water
- distinguish between hard and soft water in nature
- apply methods of softening hard water in day-to-day life

Inquiry questions:

- What is the importance of different types of water?
- Why is hard water preferred for drinking?

Before reading, write one thing you already know and one question you want this lesson to answer.

Water hardness: physical properties of water: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Water hardness: physical properties of water.

Important idea: Water hardness: physical properties of water.

Water chemistry explains properties such as hardness, solubility, salts, soaps, and detergents. A strong answer names the ions or substances involved, describes the observation, and explains the chemical reason for the result.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Water hardness: physical properties of water
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Water hardness: physical properties of water
- Water hardness
- Water
- hardness
- physical
- properties

Explain the idea in your own words before attempting the activities.

Water hardness: physical properties of water: Worked Example

Investigation model for Water hardness: physical properties of water: Explain a water or salt property using evidence.

Question: How can an observation show the chemical property being studied?

Example: Hard water does not lather easily with soap because calcium or magnesium ions react with soap to form scum. A soapless detergent can lather better in hard water because it does not form the same scum.

Procedure:

1. State the property being tested, such as hardness, solubility, or detergent action.
2. Name the substances or ions involved where they are known.
3. Record the observation, such as lather, precipitate, residue, or solubility.
4. Explain the observation using the particles or ions.
5. Add one use, problem, or treatment method linked to the property.

Conclusion frame: The observation shows _____ because _____.

Water hardness: physical properties of water: Activity

Activity for Water hardness: physical properties of water: Plan a safe observation or investigation.

Inquiry question: What is the importance of different types of water?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Water hardness: physical properties of water: Activity And Practice

Practice for Water hardness: physical properties of water:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: describe the physical properties of water. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: distinguish between hard and soft water in nature. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: apply methods of softening hard water in day-to-day life. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Water hardness: physical properties of water: Outcome Check

Outcome: describe the physical properties of water

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Water hardness: physical properties of water, uses correct science vocabulary, and connects evidence to explanation.

Water hardness: physical properties of water: Outcome Check

Outcome: distinguish between hard and soft water in nature

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Water hardness: physical properties of water, uses correct science vocabulary, and connects evidence to explanation.

Water hardness: physical properties of water: Outcome Check

Outcome: apply methods of softening hard water in day-to-day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Water hardness: physical properties of water, uses correct science vocabulary, and connects evidence to explanation.

Water hardness: outline the advantages and disadvantages of hard: Lesson Opener

Learning area: Water hardness: outline the advantages and disadvantages of hard

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Water hardness: outline the advantages and disadvantages of hard.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Water hardness: outline the advantages and disadvantages of hard.

By the end of this lesson sequence, you should be able to:

- outline the advantages and disadvantages of hard and soft water
- appreciate the applications of soft and hard water in day-to-day life.

Inquiry questions:

- Why is hard water preferred for drinking?

Before reading, write one thing you already know and one question you want this lesson to answer.

Water hardness: outline the advantages and disadvantages of hard: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Water hardness: outline the advantages and disadvantages of hard.

Important idea: Water hardness: outline the advantages and disadvantages of hard.

Water chemistry explains properties such as hardness, solubility, salts, soaps, and detergents. A strong answer names the ions or substances involved, describes the observation, and explains the chemical reason for the result.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Water hardness: outline the advantages and disadvantages of hard
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Water hardness: outline the advantages and disadvantages of hard
- Water hardness
- Water
- hardness
- outline
- advantages
- disadvantages

Explain the idea in your own words before attempting the activities.

Water hardness: outline the advantages and disadvantages of hard: Worked Example

Investigation model for Water hardness: outline the advantages and disadvantages of hard: Explain a water or salt property using evidence.

Question: How can an observation show the chemical property being studied?

Example: Hard water does not lather easily with soap because calcium or magnesium ions react with soap to form scum. A soapless detergent can lather better in hard water because it does not form the same scum.

Procedure:

1. State the property being tested, such as hardness, solubility, or detergent action.
2. Name the substances or ions involved where they are known.
3. Record the observation, such as lather, precipitate, residue, or solubility.
4. Explain the observation using the particles or ions.
5. Add one use, problem, or treatment method linked to the property.

Conclusion frame: The observation shows _____ because _____.

Water hardness: outline the advantages and disadvantages of hard: Activity

Activity for Water hardness: outline the advantages and disadvantages of hard: Plan a safe observation or investigation.

Inquiry question: Why is hard water preferred for drinking?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Water hardness: outline the advantages and disadvantages of hard: Activity And Practice

Practice for Water hardness: outline the advantages and disadvantages of hard:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: outline the advantages and disadvantages of hard and soft water. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: appreciate the applications of soft and hard water in day-to-day life.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Water hardness: outline the advantages and disadvantages of hard: Outcome Check

Outcome: outline the advantages and disadvantages of hard and soft water

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Water hardness: outline the advantages and disadvantages of hard, uses correct science vocabulary, and connects evidence to explanation.

Water hardness: outline the advantages and disadvantages of hard: Outcome Check

Outcome: appreciate the applications of soft and hard water in day-to-day life.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Water hardness: outline the advantages and disadvantages of hard, uses correct science vocabulary, and connects evidence to explanation.

Mixtures, Elements, and Compounds: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Living things and their Environment

In this chapter you will study Living things and their Environment.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 2.1 Nutrition in Plants
- 2.2 Nutrition in animals
- 2.3 Reproduction in plants
- 2.4 The Interdependence of Life

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Nutrition in Plants: external and internal parts of a leaf: Lesson Opener

Learning area: Nutrition in Plants: external and internal parts of a leaf

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Nutrition in Plants: external and internal parts of a leaf.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Nutrition in Plants: external and internal parts of a leaf.

By the end of this lesson sequence, you should be able to:

- identify external and internal parts of a leaf
- explain adaptations of the leaf to photosynthesis
- describe the process of photosynthesis

Inquiry questions:

- What is the importance of photosynthesis in nature?

Before reading, write one thing you already know and one question you want this lesson to answer.

Nutrition in Plants: external and internal parts of a leaf: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Nutrition in Plants: external and internal parts of a leaf.

Important idea: Nutrition in Plants: external and internal parts of a leaf.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Nutrition in Plants: external and internal parts of a leaf
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Nutrition in Plants: external and internal parts of a leaf
- Nutrition in Plants
- Nutrition
- Plants
- external
- internal
- parts

Explain the idea in your own words before attempting the activities.

Nutrition in Plants: external and internal parts of a leaf: Worked Example

Observation model for Nutrition in Plants: external and internal parts of a leaf: Link plant parts to functions.

Question: How does each plant part help the plant live?

Example: Roots hold the plant in soil and absorb water. The stem supports leaves and carries water upward. Leaves receive light and make food. Flowers help the plant produce seeds.

Procedure:

1. Draw a plant with roots, stem, leaves, and flower where present.
2. Label each part clearly.
3. Write one function next to each label.
4. Add arrows to show water movement from root to stem to leaves.
5. Explain one adaptation, such as broad leaves for light or deep roots for water.

Conclusion frame: A plant survives because its parts _____.

Nutrition in Plants: external and internal parts of a leaf: Activity

Activity for Nutrition in Plants: external and internal parts of a leaf: Plan a safe observation or investigation.

Inquiry question: What is the importance of photosynthesis in nature?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Nutrition in Plants: external and internal parts of a leaf: Activity And Practice

Practice for Nutrition in Plants: external and internal parts of a leaf:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: identify external and internal parts of a leaf. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: explain adaptations of the leaf to photosynthesis. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: describe the process of photosynthesis. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Nutrition in Plants: external and internal parts of a leaf: Outcome Check

Outcome: identify external and internal parts of a leaf

Show mastery with topic evidence.

Task: Draw a labelled plant and explain the function of root, stem, leaf, and flower. Answer guide: roots absorb water and anchor the plant, stem supports and transports, leaves make food, flowers support reproduction.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Nutrition in Plants: external and internal parts of a leaf, uses correct science vocabulary, and connects evidence to explanation.

Nutrition in Plants: external and internal parts of a leaf: Outcome Check

Outcome: explain adaptations of the leaf to photosynthesis

Show mastery with topic evidence.

Task: Draw a labelled plant and explain the function of root, stem, leaf, and flower. Answer guide: roots absorb water and anchor the plant, stem supports and transports, leaves make food, flowers support reproduction.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Nutrition in Plants: external and internal parts of a leaf, uses correct science vocabulary, and connects evidence to explanation.

Nutrition in Plants: external and internal parts of a leaf: Outcome Check

Outcome: describe the process of photosynthesis

Show mastery with topic evidence.

Task: Draw a labelled plant and explain the function of root, stem, leaf, and flower. Answer guide: roots absorb water and anchor the plant, stem supports and transports, leaves make food, flowers support reproduction.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Nutrition in Plants: external and internal parts of a leaf, uses correct science vocabulary, and connects evidence to explanation.

Nutrition in Plants: investigate the conditions necessary for photosynthesis: Lesson Opener

Learning area: Nutrition in Plants: investigate the conditions necessary for photosynthesis

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Nutrition in Plants: investigate the conditions necessary for photosynthesis.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Nutrition in Plants: investigate the conditions necessary for photosynthesis.

By the end of this lesson sequence, you should be able to:

- investigate the conditions necessary for photosynthesis
- appreciate the process of photosynthesis in nature.

Inquiry questions:

- What is the importance of photosynthesis in nature?

Before reading, write one thing you already know and one question you want this lesson to answer.

Nutrition in Plants: investigate the conditions necessary for photosynthesis: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Nutrition in Plants: investigate the conditions necessary for photosynthesis.

Important idea: Nutrition in Plants: investigate the conditions necessary for photosynthesis.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Nutrition in Plants: investigate the conditions necessary for photosynthesis |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Nutrition in Plants: investigate the conditions necessary for photosynthesis
- Nutrition in Plants
- Nutrition
- Plants
- investigate
- conditions
- necessary
- photosynthesis

Explain the idea in your own words before attempting the activities.

Nutrition in Plants: investigate the conditions necessary for photosynthesis: Worked Example

Investigation model for Nutrition in Plants: investigate the conditions necessary for photosynthesis: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Nutrition in Plants: investigate the conditions necessary for photosynthesis: Activity

Activity for Nutrition in Plants: investigate the conditions necessary for photosynthesis: Plan a safe observation or investigation.

Inquiry question: What is the importance of photosynthesis in nature?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Nutrition in Plants: investigate the conditions necessary for photosynthesis: Activity And Practice

Practice for Nutrition in Plants: investigate the conditions necessary for photosynthesis:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: investigate the conditions necessary for photosynthesis. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: appreciate the process of photosynthesis in nature.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Nutrition in Plants: investigate the conditions necessary for photosynthesis: Outcome Check

Outcome: investigate the conditions necessary for photosynthesis

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Nutrition in Plants: investigate the conditions necessary for photosynthesis, uses correct science vocabulary, and connects evidence to explanation.

Nutrition in Plants: investigate the conditions necessary for photosynthesis: Outcome Check

Outcome: appreciate the process of photosynthesis in nature.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Nutrition in Plants: investigate the conditions necessary for photosynthesis, uses correct science vocabulary, and connects evidence to explanation.

Nutrition in animals: outline modes of nutrition in animals: Lesson Opener

Learning area: Nutrition in animals: outline modes of nutrition in animals

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Nutrition in animals: outline modes of nutrition in animals.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Nutrition in animals: outline modes of nutrition in animals.

By the end of this lesson sequence, you should be able to:

- outline modes of nutrition in animals
- describe the structure and functions of different types of teeth
- classify animals based on their dentition

Inquiry questions:

- How do different animals feed?
- How is food digested in the human body?

Before reading, write one thing you already know and one question you want this lesson to answer.

Nutrition in animals: outline modes of nutrition in animals: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Nutrition in animals: outline modes of nutrition in animals.

Important idea: Nutrition in animals: outline modes of nutrition in animals.

Digestion changes food into small soluble substances the body can absorb. Mechanical digestion breaks food into smaller pieces. Chemical digestion uses enzymes to break large food molecules into smaller ones. A good explanation follows food through the mouth, oesophagus, stomach, small intestine, large intestine, and rectum.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Nutrition in animals: outline modes of nutrition in animals
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Nutrition in animals: outline modes of nutrition in animals
- Nutrition in animals
- Nutrition
- animals
- outline
- modes

Explain the idea in your own words before attempting the activities.

Nutrition in animals: outline modes of nutrition in animals: Worked Example

Investigation model for Nutrition in animals: outline modes of nutrition in animals: Trace what happens to food in the human digestive system.

Question: How do mechanical and chemical digestion work together?

Example: In the mouth, teeth break food into smaller pieces while saliva starts chemical digestion of starch. In the stomach, muscles churn food and gastric juice begins protein digestion. In the small intestine, enzymes complete digestion and digested food is absorbed.

Procedure:

1. Draw a labelled path: mouth, oesophagus, stomach, small intestine, large intestine, rectum.
2. Beside each part, write whether the main action is mechanical digestion, chemical digestion, absorption, or removal of waste.
3. Use arrows to show movement of food.
4. Add one enzyme or digestive juice where it is taught in class.
5. Explain why digestion is not complete in the mouth only.

Conclusion frame: Digestion needs both physical breakdown and chemical breakdown because _____.

Nutrition in animals: outline modes of nutrition in animals: Activity

Activity for Nutrition in animals: outline modes of nutrition in animals: Plan a safe observation or investigation.

Inquiry question: How do different animals feed?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Nutrition in animals: outline modes of nutrition in animals: Activity And Practice

Practice for Nutrition in animals: outline modes of nutrition in animals:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: outline modes of nutrition in animals. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: describe the structure and functions of different types of teeth. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: classify animals based on their dentition. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Nutrition in animals: outline modes of nutrition in animals: Outcome Check

Outcome: outline modes of nutrition in animals

Show mastery with topic evidence.

Task: Trace the food path through the digestive system, name mechanical and chemical digestion steps, and explain one adaptation of a digestive organ.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Nutrition in animals: outline modes of nutrition in animals, uses correct science vocabulary, and connects evidence to explanation.

Nutrition in animals: outline modes of nutrition in animals: Outcome Check

Outcome: describe the structure and functions of different types of teeth

Show mastery with topic evidence.

Task: Trace the food path through the digestive system, name mechanical and chemical digestion steps, and explain one adaptation of a digestive organ.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Nutrition in animals: outline modes of nutrition in animals, uses correct science vocabulary, and connects evidence to explanation.

Nutrition in animals: outline modes of nutrition in animals: Outcome Check

Outcome: classify animals based on their dentition

Show mastery with topic evidence.

Task: Trace the food path through the digestive system, name mechanical and chemical digestion steps, and explain one adaptation of a digestive organ.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Nutrition in animals: outline modes of nutrition in animals, uses correct science vocabulary, and connects evidence to explanation.

Nutrition in animals: process of digestion in human beings: Lesson Opener

Learning area: Nutrition in animals: process of digestion in human beings

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Nutrition in animals: process of digestion in human beings.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Nutrition in animals: process of digestion in human beings.

By the end of this lesson sequence, you should be able to:

- describe the process of digestion in human beings
- appreciate that animals have varied modes of nutrition.

Inquiry questions:

- How is food digested in the human body?

Before reading, write one thing you already know and one question you want this lesson to answer.

Nutrition in animals: process of digestion in human beings: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Nutrition in animals: process of digestion in human beings.

Important idea: Nutrition in animals: process of digestion in human beings.

Digestion changes food into small soluble substances the body can absorb. Mechanical digestion breaks food into smaller pieces. Chemical digestion uses enzymes to break large food molecules into smaller ones. A good explanation follows food through the mouth, oesophagus, stomach, small intestine, large intestine, and rectum.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Nutrition in animals: process of digestion in human beings |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Nutrition in animals: process of digestion in human beings
- Nutrition in animals
- Nutrition
- animals
- process
- digestion
- human
- beings

Explain the idea in your own words before attempting the activities.

Nutrition in animals: process of digestion in human beings:

Worked Example

Investigation model for Nutrition in animals: process of digestion in human beings: Trace what happens to food in the human digestive system.

Question: How do mechanical and chemical digestion work together?

Example: In the mouth, teeth break food into smaller pieces while saliva starts chemical digestion of starch. In the stomach, muscles churn food and gastric juice begins protein digestion. In the small intestine, enzymes complete digestion and digested food is absorbed.

Procedure:

1. Draw a labelled path: mouth, oesophagus, stomach, small intestine, large intestine, rectum.
2. Beside each part, write whether the main action is mechanical digestion, chemical digestion, absorption, or removal of waste.
3. Use arrows to show movement of food.
4. Add one enzyme or digestive juice where it is taught in class.
5. Explain why digestion is not complete in the mouth only.

Conclusion frame: Digestion needs both physical breakdown and chemical breakdown because _____.

Nutrition in animals: process of digestion in human beings:

Activity

Activity for Nutrition in animals: process of digestion in human beings: Plan a safe observation or investigation.

Inquiry question: How is food digested in the human body?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Nutrition in animals: process of digestion in human beings:

Activity And Practice

Practice for Nutrition in animals: process of digestion in human beings:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: describe the process of digestion in human beings. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: appreciate that animals have varied modes of nutrition.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Nutrition in animals: process of digestion in human beings: Outcome Check

Outcome: describe the process of digestion in human beings

Show mastery with topic evidence.

Task: Trace the food path through the digestive system, name mechanical and chemical digestion steps, and explain one adaptation of a digestive organ.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Nutrition in animals: process of digestion in human beings, uses correct science vocabulary, and connects evidence to explanation.

Nutrition in animals: process of digestion in human beings: Outcome Check

Outcome: appreciate that animals have varied modes of nutrition.

Show mastery with topic evidence.

Task: Trace the food path through the digestive system, name mechanical and chemical digestion steps, and explain one adaptation of a digestive organ.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Nutrition in animals: process of digestion in human beings, uses correct science vocabulary, and connects evidence to explanation.

Reproduction in plants: outline functions of parts of a flower: Lesson Opener

Learning area: Reproduction in plants: outline functions of parts of a flower

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Reproduction in plants: outline functions of parts of a flower.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Reproduction in plants: outline functions of parts of a flower.

By the end of this lesson sequence, you should be able to:

- outline functions of parts of a flower
- describe pollination in plants
- outline the adaptations of flowers to wind and insect pollination

Inquiry questions:

- How does reproduction in plants occur?

Before reading, write one thing you already know and one question you want this lesson to answer.

Reproduction in plants: outline functions of parts of a flower: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Reproduction in plants: outline functions of parts of a flower.

Important idea: Reproduction in plants: outline functions of parts of a flower.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Reproduction in plants: outline functions of parts of a flower
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Reproduction in plants: outline functions of parts of a flower
- Reproduction in plants
- Reproduction
- plants
- outline
- functions
- parts
- flower

Explain the idea in your own words before attempting the activities.

Reproduction in plants: outline functions of parts of a flower: Worked Example

Observation model for Reproduction in plants: outline functions of parts of a flower: Link plant parts to functions.

Question: How does each plant part help the plant live?

Example: Roots hold the plant in soil and absorb water. The stem supports leaves and carries water upward. Leaves receive light and make food. Flowers help the plant produce seeds.

Procedure:

1. Draw a plant with roots, stem, leaves, and flower where present.
2. Label each part clearly.
3. Write one function next to each label.
4. Add arrows to show water movement from root to stem to leaves.
5. Explain one adaptation, such as broad leaves for light or deep roots for water.

Conclusion frame: A plant survives because its parts _____.

Reproduction in plants: outline functions of parts of a flower:

Activity

Activity for Reproduction in plants: outline functions of parts of a flower: Plan a safe observation or investigation.

Inquiry question: How does reproduction in plants occur?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Reproduction in plants: outline functions of parts of a flower:

Activity And Practice

Practice for Reproduction in plants: outline functions of parts of a flower:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: outline functions of parts of a flower. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: describe pollination in plants. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: outline the adaptations of flowers to wind and insect pollination. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Reproduction in plants: outline functions of parts of a flower: Outcome Check

Outcome: outline functions of parts of a flower

Show mastery with topic evidence.

Task: Draw a labelled plant and explain the function of root, stem, leaf, and flower. Answer guide: roots absorb water and anchor the plant, stem supports and transports, leaves make food, flowers support reproduction.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Reproduction in plants: outline functions of parts of a flower, uses correct science vocabulary, and connects evidence to explanation.

Reproduction in plants: outline functions of parts of a flower: Outcome Check

Outcome: describe pollination in plants

Show mastery with topic evidence.

Task: Draw a labelled plant and explain the function of root, stem, leaf, and flower. Answer guide: roots absorb water and anchor the plant, stem supports and transports, leaves make food, flowers support reproduction.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Reproduction in plants: outline functions of parts of a flower, uses correct science vocabulary, and connects evidence to explanation.

Reproduction in plants: outline functions of parts of a flower: Outcome Check

Outcome: outline the adaptations of flowers to wind and insect pollination

Show mastery with topic evidence.

Task: Draw a labelled plant and explain the function of root, stem, leaf, and flower. Answer guide: roots absorb water and anchor the plant, stem supports and transports, leaves make food, flowers support reproduction.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Reproduction in plants: outline functions of parts of a flower, uses correct science vocabulary, and connects evidence to explanation.

Reproduction in plants: explain fertilisation and fruit formation in flowering plants: Lesson Opener

Learning area: Reproduction in plants: explain fertilisation and fruit formation in flowering plants

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Reproduction in plants: explain fertilisation and fruit formation in flowering plants.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Reproduction in plants: explain fertilisation and fruit formation in flowering plants.

By the end of this lesson sequence, you should be able to:

- explain fertilisation and fruit formation in flowering plants
- categorise fruits and seeds based on their mode of dispersal
- recognise the role of flowers in nature.

Inquiry questions:

- How does reproduction in plants occur?

Before reading, write one thing you already know and one question you want this lesson to answer.

Reproduction in plants: explain fertilisation and fruit formation in flowering plants: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Reproduction in plants: explain fertilisation and fruit formation in flowering plants.

Important idea: Reproduction in plants: explain fertilisation and fruit formation in flowering plants.

Living systems are made of parts that work together. Learn each part by linking structure to function: what it is like, what it does, and how it supports the whole organism or system. Use labelled diagrams and evidence from observation or approved class texts.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Reproduction in plants: explain fertilisation and fruit formation in flowering plants |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Reproduction in plants: explain fertilisation and fruit formation in flowering plants
- Reproduction in plants
- Reproduction
- plants
- explain
- fertilisation
- fruit
- formation

Explain the idea in your own words before attempting the activities.

Reproduction in plants: explain fertilisation and fruit formation in flowering plants: Worked Example

Investigation model for Reproduction in plants: explain fertilisation and fruit formation in flowering plants: Connect a living structure to its function.

Question: How does one part support the work of the whole system?

Example: In the breathing system, the trachea carries air, bronchi branch into the lungs, and alveoli provide a large surface for gas exchange.

Procedure:

1. Draw the organism, organ, or system being studied.
2. Label each visible part clearly.
3. For each label, write one function using the frame: "This part helps by..."
4. Connect two parts with an arrow and explain how they work together.
5. Check that every claim is based on observation, a diagram, class text, or teacher demonstration.

Conclusion frame: The evidence shows that _____ is important because _____.

Reproduction in plants: explain fertilisation and fruit formation in flowering plants: Activity

Activity for Reproduction in plants: explain fertilisation and fruit formation in flowering plants: Plan a safe observation or investigation.

Inquiry question: How does reproduction in plants occur?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Reproduction in plants: explain fertilisation and fruit formation in flowering plants: Activity And Practice

Practice for Reproduction in plants: explain fertilisation and fruit formation in flowering plants:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: explain fertilisation and fruit formation in flowering plants. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: categorise fruits and seeds based on their mode of dispersal. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: recognise the role of flowers in nature.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Reproduction in plants: explain fertilisation and fruit formation in flowering plants: Outcome Check

Outcome: explain fertilisation and fruit formation in flowering plants

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Reproduction in plants: explain fertilisation and fruit formation in flowering plants, uses correct science vocabulary, and connects evidence to explanation.

Reproduction in plants: explain fertilisation and fruit formation in flowering plants: Outcome Check

Outcome: categorise fruits and seeds based on their mode of dispersal

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Reproduction in plants: explain fertilisation and fruit formation in flowering plants, uses correct science vocabulary, and connects evidence to explanation.

Reproduction in plants: explain fertilisation and fruit formation in flowering plants: Outcome Check

Outcome: recognise the role of flowers in nature.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Reproduction in plants: explain fertilisation and fruit formation in flowering plants, uses correct science vocabulary, and connects evidence to explanation.

The Interdependence of Life: explain the biotic and abiotic factors: Lesson Opener

Learning area: The Interdependence of Life: explain the biotic and abiotic factors

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to The Interdependence of Life: explain the biotic and abiotic factors.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on The Interdependence of Life: explain the biotic and abiotic factors.

By the end of this lesson sequence, you should be able to:

- explain the biotic and abiotic factors of the environment
- construct food chains and food webs in the environment
- describe the effect of human activities on the environment

Inquiry questions:

- What is the role of living and non-living factors in environments?

Before reading, write one thing you already know and one question you want this lesson to answer.

The Interdependence of Life: explain the biotic and abiotic factors: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to The Interdependence of Life: explain the biotic and abiotic factors.

Important idea: The Interdependence of Life: explain the biotic and abiotic factors.

Environmental science connects observation with care for people and living things. Record place, date, conditions, and evidence. Then explain the pattern and propose a practical improvement that protects health, safety, food, water, or the environment.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in The Interdependence of Life: explain the biotic and abiotic factors
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- The Interdependence of Life: explain the biotic and abiotic factors
- The Interdependence of Life
- Interdependence
- Life
- explain
- biotic
- abiotic

Explain the idea in your own words before attempting the activities.

The Interdependence of Life: explain the biotic and abiotic factors: Worked Example

Investigation model for The Interdependence of Life: explain the biotic and abiotic factors: Use field observation carefully.

Question: What evidence in the local environment shows the pattern or problem?

Example: A school garden with dry soil, wilting leaves, and cracks near the roots shows that plants need steady water and soil cover to reduce moisture loss.

Procedure:

1. Choose a safe observation point in the school compound, garden, home area, or community.
2. Record date, weather, place, and what you can see without disturbing living things.
3. Draw a labelled sketch or complete a three-column observation table.
4. Separate facts from explanations.
5. Suggest one practical improvement that protects health, safety, food, water, or the environment.

Conclusion frame: My observations show _____ because the evidence was _____.

The Interdependence of Life: explain the biotic and abiotic factors: Activity

Activity for The Interdependence of Life: explain the biotic and abiotic factors: Plan a safe observation or investigation.

Inquiry question: What is the role of living and non-living factors in environments?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

The Interdependence of Life: explain the biotic and abiotic factors: Activity And Practice

Practice for The Interdependence of Life: explain the biotic and abiotic factors:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: explain the biotic and abiotic factors of the environment. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: construct food chains and food webs in the environment. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: describe the effect of human activities on the environment. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

The Interdependence of Life: explain the biotic and abiotic factors: Outcome Check

Outcome: explain the biotic and abiotic factors of the environment

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in The Interdependence of Life: explain the biotic and abiotic factors, uses correct science vocabulary, and connects evidence to explanation.

The Interdependence of Life: explain the biotic and abiotic factors: Outcome Check

Outcome: construct food chains and food webs in the environment

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in The Interdependence of Life: explain the biotic and abiotic factors, uses correct science vocabulary, and connects evidence to explanation.

The Interdependence of Life: explain the biotic and abiotic factors: Outcome Check

Outcome: describe the effect of human activities on the environment

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in The Interdependence of Life: explain the biotic and abiotic factors, uses correct science vocabulary, and connects evidence to explanation.

The Interdependence of Life: interdependence between living and non-living factors: Lesson Opener

Learning area: The Interdependence of Life: interdependence between living and non-living factors

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to The Interdependence of Life: interdependence between living and non-living factors.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on The Interdependence of Life: interdependence between living and non-living factors.

By the end of this lesson sequence, you should be able to:

- appreciate the interdependence between living and non-living factors of the environment.

Inquiry questions:

- What is the role of living and non-living factors in environments?

Before reading, write one thing you already know and one question you want this lesson to answer.

The Interdependence of Life: interdependence between living and non-living factors: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to The Interdependence of Life: interdependence between living and non-living factors.

Important idea: The Interdependence of Life: interdependence between living and non-living factors.

Environmental science connects observation with care for people and living things. Record place, date, conditions, and evidence. Then explain the pattern and propose a practical improvement that protects health, safety, food, water, or the environment.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in The Interdependence of Life: interdependence between living and non-living factors |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- The Interdependence of Life: interdependence between living and non-living factors
- The Interdependence of Life
- Interdependence
- Life
- between
- living
- non-living

Explain the idea in your own words before attempting the activities.

The Interdependence of Life: interdependence between living and non-living factors: Worked Example

Investigation model for The Interdependence of Life: interdependence between living and non-living factors: Use field observation carefully.

Question: What evidence in the local environment shows the pattern or problem?

Example: A school garden with dry soil, wilting leaves, and cracks near the roots shows that plants need steady water and soil cover to reduce moisture loss.

Procedure:

1. Choose a safe observation point in the school compound, garden, home area, or community.
2. Record date, weather, place, and what you can see without disturbing living things.
3. Draw a labelled sketch or complete a three-column observation table.
4. Separate facts from explanations.
5. Suggest one practical improvement that protects health, safety, food, water, or the environment.

Conclusion frame: My observations show _____ because the evidence was _____.

The Interdependence of Life: interdependence between living and non-living factors: Activity

Activity for The Interdependence of Life: interdependence between living and non-living factors: Plan a safe observation or investigation.

Inquiry question: What is the role of living and non-living factors in environments?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

The Interdependence of Life: interdependence between living and non-living factors: Activity And Practice

Practice for The Interdependence of Life: interdependence between living and non-living factors:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: appreciate the interdependence between living and non-living factors of the environment.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

The Interdependence of Life: interdependence between living and non-living factors: Outcome Check

Outcome: appreciate the interdependence between living and non-living factors of the environment.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in The Interdependence of Life: interdependence between living and non-living factors, uses correct science vocabulary, and connects evidence to explanation.

Living things and their Environment: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Force and Energy

In this chapter you will study Force and Energy.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 3.1 Curved mirrors
- 3.2 Waves

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Curved mirrors: types of curved mirrors: Lesson Opener

Learning area: Curved mirrors: types of curved mirrors

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Curved mirrors: types of curved mirrors.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Curved mirrors: types of curved mirrors.

By the end of this lesson sequence, you should be able to:

- describe types of curved mirrors
- draw ray diagrams to locate images formed by concave and convex mirrors
- describe the characteristics of images formed by concave and convex mirrors

Inquiry questions:

- How are curved mirrors used in day-to-day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Curved mirrors: types of curved mirrors: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Curved mirrors: types of curved mirrors.

Important idea: Curved mirrors: types of curved mirrors.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Curved mirrors: types of curved mirrors
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Curved mirrors: types of curved mirrors
- Curved mirrors
- Curved
- mirrors
- types

Explain the idea in your own words before attempting the activities.

Curved mirrors: types of curved mirrors: Worked Example

Investigation model for Curved mirrors: types of curved mirrors: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Curved mirrors: types of curved mirrors: Activity

Activity for Curved mirrors: types of curved mirrors: Plan a safe observation or investigation.

Inquiry question: How are curved mirrors used in day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Curved mirrors: types of curved mirrors: Activity And Practice

Practice for Curved mirrors: types of curved mirrors:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: describe types of curved mirrors. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: draw ray diagrams to locate images formed by concave and convex mirrors. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: describe the characteristics of images formed by concave and convex mirrors. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Curved mirrors: types of curved mirrors: Outcome Check

Outcome: describe types of curved mirrors

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Curved mirrors: types of curved mirrors, uses correct science vocabulary, and connects evidence to explanation.

Curved mirrors: types of curved mirrors: Outcome Check

Outcome: draw ray diagrams to locate images formed by concave and convex mirrors

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Curved mirrors: types of curved mirrors, uses correct science vocabulary, and connects evidence to explanation.

Curved mirrors: types of curved mirrors: Outcome Check

Outcome: describe the characteristics of images formed by concave and convex mirrors

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Curved mirrors: types of curved mirrors, uses correct science vocabulary, and connects evidence to explanation.

Curved mirrors: explain the uses of concave and convex mirrors: Lesson Opener

Learning area: Curved mirrors: explain the uses of concave and convex mirrors

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Curved mirrors: explain the uses of concave and convex mirrors.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Curved mirrors: explain the uses of concave and convex mirrors.

By the end of this lesson sequence, you should be able to:

- explain the uses of concave and convex mirrors in day-to-day life
- appreciate the applications of curved mirrors

Inquiry questions:

- How are curved mirrors used in day-to-day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Curved mirrors: explain the uses of concave and convex mirrors: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Curved mirrors: explain the uses of concave and convex mirrors.

Important idea: Curved mirrors: explain the uses of concave and convex mirrors.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Curved mirrors: explain the uses of concave and convex mirrors
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Curved mirrors: explain the uses of concave and convex mirrors
- Curved mirrors
- Curved
- mirrors
- explain
- concave
- convex

Explain the idea in your own words before attempting the activities.

Curved mirrors: explain the uses of concave and convex mirrors: Worked Example

Investigation model for Curved mirrors: explain the uses of concave and convex mirrors: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Curved mirrors: explain the uses of concave and convex mirrors: Activity

Activity for Curved mirrors: explain the uses of concave and convex mirrors: Plan a safe observation or investigation.

Inquiry question: How are curved mirrors used in day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Curved mirrors: explain the uses of concave and convex mirrors: Activity And Practice

Practice for Curved mirrors: explain the uses of concave and convex mirrors:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: explain the uses of concave and convex mirrors in day-to-day life. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: appreciate the applications of curved mirrors. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Curved mirrors: explain the uses of concave and convex mirrors: Outcome Check

Outcome: explain the uses of concave and convex mirrors in day-to-day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Curved mirrors: explain the uses of concave and convex mirrors, uses correct science vocabulary, and connects evidence to explanation.

Curved mirrors: explain the uses of concave and convex mirrors: Outcome Check

Outcome: appreciate the applications of curved mirrors

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Curved mirrors: explain the uses of concave and convex mirrors, uses correct science vocabulary, and connects evidence to explanation.

Waves: generation of waves in nature: Lesson Opener

Learning area: Waves: generation of waves in nature

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Waves: generation of waves in nature.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Waves: generation of waves in nature.

By the end of this lesson sequence, you should be able to:

- describe the generation of waves in nature
- classify waves as longitudinal and transverse, carry out activities to demonstrate the parts
- describe basic characteristics of waves in nature

Inquiry questions:

- How are waves applied in our day-to-day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Waves: generation of waves in nature: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Waves: generation of waves in nature.

Important idea: Waves: generation of waves in nature.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Waves: generation of waves in nature |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Waves: generation of waves in nature
- Waves
- generation
- nature

Explain the idea in your own words before attempting the activities.

Waves: generation of waves in nature: Worked Example

Investigation model for Waves: generation of waves in nature: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Waves: generation of waves in nature: Activity

Activity for Waves: generation of waves in nature: Plan a safe observation or investigation.

Inquiry question: How are waves applied in our day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Waves: generation of waves in nature: Activity And Practice

Practice for Waves: generation of waves in nature:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: describe the generation of waves in nature. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: classify waves as longitudinal and transverse, carry out activities to demonstrate the parts. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: describe basic characteristics of waves in nature. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Waves: generation of waves in nature: Outcome Check

Outcome: describe the generation of waves in nature

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Waves: generation of waves in nature, uses correct science vocabulary, and connects evidence to explanation.

Waves: generation of waves in nature: Outcome Check

Outcome: classify waves as longitudinal and transverse, carry out activities to demonstrate the parts

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Waves: generation of waves in nature, uses correct science vocabulary, and connects evidence to explanation.

Waves: generation of waves in nature: Outcome Check

Outcome: describe basic characteristics of waves in nature

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Waves: generation of waves in nature, uses correct science vocabulary, and connects evidence to explanation.

Waves: remote sensing in relation carry out activities: Lesson Opener

Learning area: Waves: remote sensing in relation carry out activities

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Waves: remote sensing in relation carry out activities.

Curriculum link: This lesson follows the Grade 9 curriculum sequence and focuses on Waves: remote sensing in relation carry out activities.

By the end of this lesson sequence, you should be able to:

- describe remote sensing in relation carry out activities to demonstrate to waves
- describe applications of waves in day-to- brainstorm on the meaning of remote day life
- appreciate the discuss remote sensing in relation to waves, applications

Inquiry questions:

- How are waves applied in our day-to- day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Waves: remote sensing in relation carry out activities: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Waves: remote sensing in relation carry out activities.

Important idea: Waves: remote sensing in relation carry out activities.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 9, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Waves: remote sensing in relation carry out activities |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Waves: remote sensing in relation carry out activities
- Waves
- remote
- sensing
- relation
- carry
- activities

Explain the idea in your own words before attempting the activities.

Waves: remote sensing in relation carry out activities: Worked Example

Investigation model for Waves: remote sensing in relation carry out activities: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Waves: remote sensing in relation carry out activities: Activity

Activity for Waves: remote sensing in relation carry out activities: Plan a safe observation or investigation.

Inquiry question: How are waves applied in our day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Waves: remote sensing in relation carry out activities: Activity And Practice

Practice for Waves: remote sensing in relation carry out activities:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: describe remote sensing in relation carry out activities to demonstrate to waves. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: describe applications of waves in day-to-day brainstorm on the meaning of remote day life. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: appreciate the discuss remote sensing in relation to waves, applications. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Waves: remote sensing in relation carry out activities: Outcome Check

Outcome: describe remote sensing in relation carry out activities to demonstrate to waves

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Waves: remote sensing in relation carry out activities, uses correct science vocabulary, and connects evidence to explanation.

Waves: remote sensing in relation carry out activities: Outcome Check

Outcome: describe applications of waves in day-to- brainstorm on the meaning of remote day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Waves: remote sensing in relation carry out activities, uses correct science vocabulary, and connects evidence to explanation.

Waves: remote sensing in relation carry out activities: Outcome Check

Outcome: appreciate the discuss remote sensing in relation to waves, applications

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Waves: remote sensing in relation carry out activities, uses correct science vocabulary, and connects evidence to explanation.

Force and Energy: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Glossary

Use this glossary to revise important words in Science and Technology.

- Structure of the atom: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Metals and Alloys: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Water hardness: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Nutrition in Plants: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Nutrition in animals: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Reproduction in plants: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- The Interdependence of Life: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Curved mirrors: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Waves: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.

Add five more words from your lessons, then write a meaning and one correct example sentence or worked example for each word.

Answer And Teacher Notes

A strong Science and Technology answer should:

- answer the exact question asked;
- separate observation from explanation;
- name materials, variables, safety checks, or data sources where needed;
- use labelled diagrams, tables, or step-by-step procedures when useful;
- draw a conclusion from evidence, not from guessing.

For investigations, good answers include a question, method, results, conclusion, and one limit or improvement. For technology tasks, good answers explain the design choice and how it solves the problem safely.

Answer-key guide: a complete science answer names the concept, shows evidence in a labelled sketch/table/model, explains the pattern, includes safety where relevant, and ends with a conclusion beginning from the evidence. Wrong answers often mix observation with opinion or leave out variables.

Rubric for self-checking:

- 4: The answer addresses the exact question, uses subject vocabulary correctly, gives evidence or working, and includes a useful check or correction.
- 3: The main answer is correct but one part needs stronger evidence, clearer working, better labels, or a fuller explanation.
- 2: Some understanding is shown, but the answer is incomplete, too general, or hard to follow.
- 1: The answer is guessed, off topic, or missing the evidence needed for the task.

Correction routine: reread the command word, underline the key data or evidence, improve one weak step, and write one sentence explaining why the final answer is reasonable.

Final Project

Create a final Science and Technology project.

Your project must include:

1. A clear question, problem, or design need.
2. Evidence from at least three lessons.
3. A safe method, labelled diagram, data table, flow chart, model, or prototype plan.
4. A conclusion that explains what the evidence shows.
5. One safety note, one limitation, and one improvement.

Present your project to a classmate, teacher, parent, or study group.